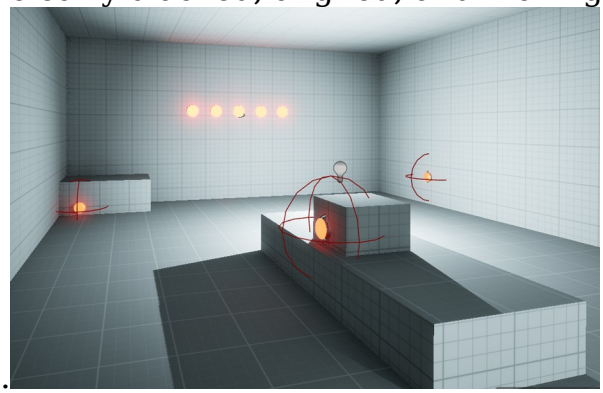
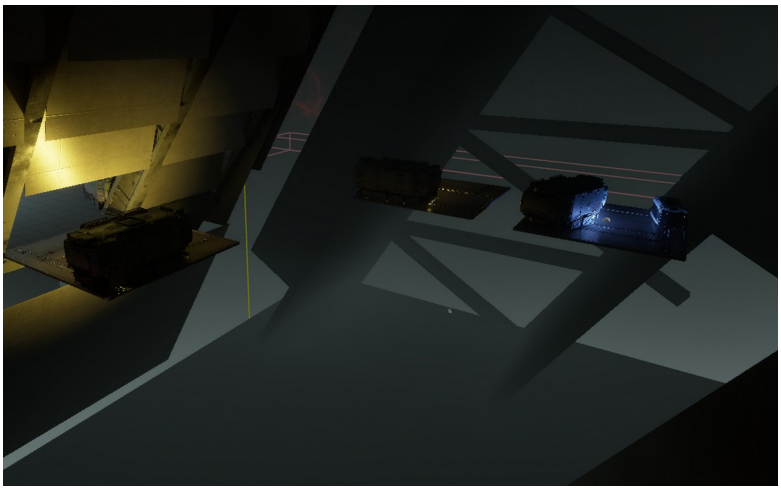
Hart, Noah - u1537288 - Actively Contributing

Over the past two sprints, I focused on bringing the train station experience and core gameplay loop together. I completed all puzzles, mechanics, and the overall design and box out of the station, while also getting the full gameplay loop functional in-editor. There are still a few issues in the build that we're actively resolving in this final week, but the core experience is now fully in place.

In addition, I set up the tutorial, helped solidify encounter design, contributed to general UI, and worked through a wide range of bugs and polish that naturally come up when systems begin to integrate. I also organized and led several engineering-focused meetings to ensure tasks were clearly tracked, aligned, and moving forward efficiently across the team.









```
public:
    ATrainStationManager();

    UPROPERTY(EditAnywhere, BlueprintReadOnly, Category = "Station")
    Changed in 0 Blueprints
    float StationTotalTimeSeconds = 480.f; // 8 minutes

    UPROPERTY(EditAnywhere, BlueprintReadOnly, Category = "Station")
    Changed in 0 Blueprints
    float ReEntryLockoutSeconds = 60.f; // 1 minute

    UPROPERTY(EditAnywhere, BlueprintReadOnly, Category = "Station")
    Changed in 0 Blueprints
    float DeathAfterDepartureDelaySeconds = 10.f;

    UPROPERTY(BlueprintReadOnly, Category = "Station")
    Changed in 0 Blueprints
    bool bStationActive = false;

    UPROPERTY(BlueprintReadOnly, Category = "Station")
    Changed in 0 Blueprints
    bool bTrainDeparted = false;

    UPROPERTY(BlueprintReadOnly, Category = "Station")

    float StationTimeRemaining = 0.f;

    UPROPERTY(BlueprintReadOnly, Category = "Station")
    Changed in 0 Blueprints
    float LockoutTimeRemaining = 0.f;
```

```
protected:
    virtual void BeginPlay() override;

    UPROPERTY(EditAnywhere, BlueprintReadOnly, Category = "Tutorial Encounter")
    Changed in 0 Blueprints
    int32 TotalEnemiesToSpawn = 8;

    UPROPERTY(EditAnywhere, BlueprintReadOnly, Category = "Tutorial Encounter")
    Changed in 0 Blueprints
    int32 EnemiesPerSpawn = 1;

    UPROPERTY(EditAnywhere, BlueprintReadOnly, Category = "Tutorial Encounter")
    float SpawnInterval = 5.0f;

    UPROPERTY(EditAnywhere, BlueprintReadOnly, Category = "Tutorial Encounter")
    FName ExitDoorTag = TEXT("Tutorial_Exit");

    UPROPERTY()
```

Engineering

For my Engineering Task, I continued refining Encounters(manager), the Trainstation manager and mechanics, Tutorial manager. These components overall contributed to the overall gameplay loop, making sure that the flow of start/(tutorial or spawncar)->train phase->station phase->repeat…Death->spawncar all function properly and cohesively. This

required testing and making sure that our specified encounter conditions were met when you won or died. The major task I completed:

Train Cars

- Redoing the Survive Wave Encounter to allow for infinite spawning after a set time limit
- Make our new doors open with an animation
- Tutorial and refactoring an old encounter to be a tutorial encounter
- Fixing bugs related to each encounter, like the capture zone not progressing when spawning in the zone, and enemies not doing damage to the defense tower

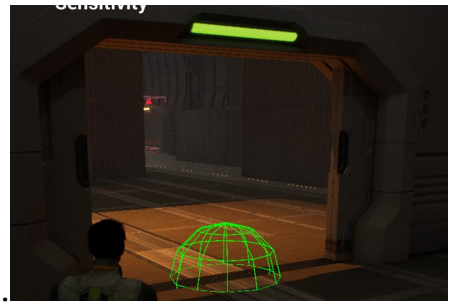
Train Station

- Complete and test the manager to give them 8 minutes to explore, otherwise they will die. Have a short lockout and have the UI change when they can go on the train car
- Elevators as part of the jumping puzzle
- A main elevator that gets unlocked by completing the paths
- Lights Out Puzzle, Drone Puzzle, and Combination Puzzle
- Implemented interactable/collectable audio logs

Design

For my design task, I dedicated the sprint following GDC to developing and boxing out the train station hub. The goal was to create a space that initially feels grounded and straightforward, but gradually reveals deeper layers as the player revisits it. With each return, players uncover hidden areas, puzzle hints, and small environmental challenges, ranging from navigation and traversal to light platforming, adding variety and encouraging continued exploration.

Because the station serves as a recurring hub between encounters, I wanted it to balance pressure and reward. Players should feel a sense of urgency as they prepare for the next challenge with a time limit of the train departing, while also being consistently rewarded for engaging with the space and discovering new elements. The design aims to ensure that even within a relatively compact environment, players always have something



new to explore beyond the main paths.

